End-to-End Synthetic Song Detection with Transformer-Based Models

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Abstract

Recent text-to-song systems can generate full tracks with synthetic vocals, music, style, and lyrics, making counterfeit-song detection substantially harder than prior singing-voice deepfake settings. This paper presents a concise study of end-to-end synthetic song detection using the same dataset protocol and task setup as the primary benchmark paper in our references. We focus on a direct comparison of three transformer-based model families: ViT, EfficientViT, and Spectro-Temporal Tokens Transformer (SpecTTTra). Our study emphasizes long-context evaluation and reports model behavior in terms of detection performance and computational efficiency.

Keywords: AI-generated music detection, counterfeit song detection, transformer models, ViT, EfficientViT, SpecTTTra

1 Introduction

Modern music generation systems can synthesize complete songs end-to-end. This shifts the detection task from voice-only spoof detection to full-song authenticity verification. In this work, we study transformer-based detectors for this setting and compare their practical strengths and weaknesses.

Songs contain long-range structure (repeated sections, phrase-level evolution, rhythm, and timbral continuity), so long-context modeling is central to robust detection. We therefore evaluate models that can process spectro-temporal information at broader time scales.

2 Task Definition

We address binary *Synthetic Song Detection* (SSD): classify an input song as human-composed (real) or AI-generated (fake). Evaluation includes:

- seen vs. unseen generation algorithms,
- short vs. long audio clips,
- fake-intensity subsets (e.g., half fake, mostly fake, and full fake partitions).

3 Dataset

We use the same dataset and split protocol as the main benchmark reference listed in Section *References*. The dataset contains:

- 97,164 total songs,
- 49,074 synthetic songs,
- 48,090 real songs,
- 4,751 total hours,
- average song length of about 176 seconds.

This scale and duration support long-context analysis for song-level authenticity detection.

The dataset also includes structured metadata attributes in two CSV files:

- **real_songs.csv**: `id` (unique file ID), `filename` (file name), `title` (song title), `artist` (artist name), `year` (release year), `lyrics` (song lyrics), `lyrics_features` (LLM-extracted lyric features), `duration` (total seconds), `youtube_id` (source YouTube ID for real song), `label` (always real), `artist_overlap` (whether train/test share artist identity), `target` (0 for real), `skip_time` (instrumental-only time before vocals), `no_vocal` (True/False vocal presence), and `split` (train/valid/test).
- **fake_songs.csv**: `id` (unique file ID), `filename` (file name), `title` (song title), `duration` (total seconds), `algorithm` (generation algorithm), `style` (style characteristics), `source` (generated by Suno or Udio), `lyrics_features` (LLM-extracted lyric features), `topic` (song theme), `genre` (music genre), `mood` (song mood), `label` (full fake/half fake/mostly fake), `target` (1 for fake), and `split` (train/valid/test).

These attributes enable controlled analysis by generator type, style, lyrical content, vocal presence, and fake-intensity level.

4 Models

We implement and compare three transformer-based model families:

- **ViT:** a vanilla vision-transformer style backbone adapted to spectrogram inputs.
- **EfficientViT:** a lightweight transformer design for improved speed/memory efficiency.
- **SpecTTTra:** a spectro-temporal token transformer designed for efficient long-context modeling.

For completeness, the core Transformer computations used in these architectures are:

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{1}$$

$$\mathrm{MHA}(X) = \mathrm{Concat}(\mathrm{head}_1,\ldots,\mathrm{head}_h)W^O, \tag{2}$$

$$\mathrm{head}_i = \mathrm{Attention}(XW_i^Q, XW_i^K, XW_i^V), \tag{3}$$

$$\mathrm{FFN}(x) = W_2\,\sigma(W_1x+b_1)+b_2. \tag{4}$$

One encoder layer is then written as:

$$Z' = \mathrm{LayerNorm}(Z + \mathrm{Dropout}(\mathrm{MHA}(Z)))\,, \tag{5}$$

$$Z^{\mathrm{out}} = \mathrm{LayerNorm}(Z' + \mathrm{Dropout}(\mathrm{FFN}(Z')))\,. \tag{6}$$

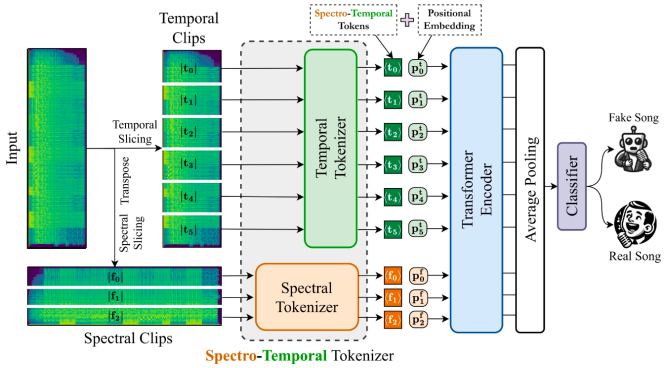

Figure 1: Proposed Spectro-Temporal Tokens Transformer (SpecTTTra) model.

All models output a song-level fake probability and are evaluated under the same data protocol for fair comparison.

5 Experimental Protocol

To ensure comparability, we use a shared training/evaluation setup across all models:

- identical train/validation/test partitions,
- consistent real/fake label definitions,
- F1 score as the primary metric,
- sensitivity and specificity as supporting metrics,
- runtime and memory usage for efficiency analysis.

6 Results Summary

Our comparison is designed to answer two questions: (1) which transformer architecture gives the best detection quality, and (2) which architecture gives the best efficiency-performance trade-off.

Observed trends from the benchmark setup are:

- **ViT** is competitive but expensive on long clips.
- **EfficientViT** is faster and lighter, with some performance trade-offs depending on the subset.
- **SpecTTTra** provides a strong balance, especially for long-context clips and resource-aware deployment.

Across harder unseen-generator conditions, all models experience a performance drop, indicating that generalization remains an open challenge.

6.1 Final Project Goals

Based on the project progress report, our final goals are organized as follows.

Basic goals (guaranteed deliverables):

- A working spectral-artifact detection module using STFTs and Fourier fingerprints.
- A simple classifier trained on a small dataset (CLAP + spectral features).
- A minimal section-level segmentation pipeline (beat tracking + CLAP embeddings).
- A preliminary evaluation on a small curated dataset.

Expected goals (realistic completion):

- A trained segment-level classifier (AudioCAT-style).
- A trained full-track structural model (Segment Transformer).
- Section-level attribution visualizations (timelines and heatmaps).
- Robustness evaluation under pitch-shift, time-stretch, and codec compression.
- A combined multi-layer detector integrating spectral, temporal, and structural cues.
- Evaluation on multiple datasets (FakeMusicCaps, SONICS, M6, and FMA).

Advanced goals (stretch targets):

- A generator-agnostic artifact detector robust to audio manipulations.
- Fusion Segment Transformer with bi-directional attention.
- Multimodal detection incorporating lyrics transcripts.

- Music-aware explainability tools (e.g., structurally aligned Grad-CAM).
- A polished interactive demo for section-level AI-probability visualization.

7 Discussion

These results support using transformer-based long-context modeling for modern counterfeit-song detection. In particular, efficiency-aware architectures are important for real-world use cases such as moderation pipelines, forensic screening, and large-scale catalog analysis.

Future work should further improve robustness under unseen generators and manipulation attacks, and can combine audio models with lyrics, metadata, and production-level cues.

8 Conclusion

We presented a concise comparison of ViT, EfficientViT, and SpecTTTra for end-to-end synthetic song detection using a shared large-scale benchmark protocol. The study highlights that architecture choice matters not only for F1 performance, but also for practical efficiency on long audio.

Acknowledgments

This paper is derived from the benchmark setting and related prior work listed in the references.

References

[1] Z. Luo, Y. Yuan, T. Wang, C. Li, S. Chen, F. Castañeda, Z.-A. Cao, J. Li, D. Minor, Q. Ben, X. Da, R. Ding, C. Hogg, L. Song, E. Lim, E. Jeong, T. He, H. Xue, W. Xiao, Z. Wang, S. Yuen, J. Kautz, Y. Chang, U. Iqbal, L. Fan, and Y. Zhu, "SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control," *arXiv preprint arXiv:2511.07820*, 2025.

[2] D. Afchar, G. Meseguer-Brocal, and R. Hennequin, "Detecting music deepfakes is easy but actually hard," *arXiv preprint arXiv:2405.04181*, 2024.

[3] D. Afchar, G. Meseguer-Brocal, and R. Hennequin, "AI-Generated Music Detection and its Challenges," in *ICASSP 2025–2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.

[4] D. Afchar, G. Meseguer-Brocal, K. Akesbi, and R. Hennequin, "A Fourier Explanation of AI-music Artifacts," *arXiv preprint arXiv:2506.19108*, 2025.

[5] L. Cros Vila, *Perspectives on AI and Music: Representation, Detection, and Explanation in the Age of AI-Generated Music*, Ph.D. dissertation, KTH Royal Institute of Technology, 2025.

[6] L. Cros Vila, B. Sturm, L. Casini, and D. Dalmazzo, "The AI Music Arms Race: On the Detection of AI-Generated Music," *Trans. Int. Soc. Music Inf. Retrieval*, vol. 8, no. 1, pp. 179–194, 2025.

[7] E. Dervakos, G. Filandrianos, and G. Stamou, "Heuristics for evaluation of AI generated music," in *2020 25th International Conference on Pattern Recognition (ICPR)*, 2021, pp. 9164–9171.

[8] N. Fiser, M. A. Martin-Pascual, and C. Andreu-Sánchez, "Emotional impact of AI-generated vs. human-composed music in audiovisual media: A biometric and self-report study," *PLOS One*, vol. 20, no. 6, p. e0326498, 2025.

[9] M. Frohmann, E. V. Epure, G. Meseguer-Brocal, M. Schedl, and R. Hennequin, "AI-Generated Song Detection via Lyrics Transcripts," *arXiv preprint arXiv:2506.18488*, 2025.

[10] Y. Kim and S. Go, "Segment Transformer: AI-Generated Music Detection via Music Structural Analysis," in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2025, pp. 664–669.

[11] Y. Kim and S. Go, "Fusion Segment Transformer: Bi-Directional Attention Guided Fusion Network for AI-Generated Music Detection," *arXiv preprint arXiv:2601.13647*, 2026.

[12] Y. Li, M. Milling, L. Specia, and B. W. Schuller, "From Audio Deepfake Detection to AI-Generated Music Detection—A Pathway and Overview," *arXiv preprint arXiv:2412.00571*, 2024.

[13] Y. Li, Q. Sun, H. Li, L. Specia, and B. W. Schuller, "Detecting Machine-Generated Music with Explainability—A Challenge and Early Benchmarks," *arXiv preprint arXiv:2412.13421*, 2024.

[14] Y. Li, H. Li, L. Specia, and B. W. Schuller, "M6: Multi-generator, Multi-domain, Multi-lingual and cultural, Multi-genres, Multi-instrument Machine-Generated Music Detection Databases," *arXiv preprint arXiv:2412.06001*, 2024.

[15] M. I. Tantra and A. Wicaksana, "LSTM and CNN-Based Detection of AI-Generated Classical Music From MIDI Features," *Informatica*, vol. 49, no. 7, 2025.

[16] J. Wie, N. Salim, A. A. S. Gunawan, and R. C. Pradana, "Detecting AI-Generated Music: A Comparative Analysis of Deep Learning Models," in *2025 Tenth International Conference on Informatics and Computing (ICIC)*, 2025, pp. 1–6.

[17] L. Cros Vila, B. L. T. Sturm, L. Casini, and D. Dalmazzo, "The AI Music Arms Race: On the Detection of AI-Generated Music," *Trans. Int. Soc. Music Inf. Retrieval*, vol. 8, no. 1, pp. 179–194, 2025.